

Understanding the Adversarial Landscape of Large Language Models Through the Lens of Attack Objectives

Nan Wang^{ID} | and Kane Walter^{ID} | CSIRO's Data61
 Yansong Gao^{ID} | The University of Western Australia
 Alsharif Abuadbba^{ID} | CSIRO's Data61

Large language models have revolutionized artificial intelligence but face growing adversarial threats. This article presents the adversarial landscape from a novel perspective by categorizing threats into four key objectives: privacy breaches, integrity compromises, adversarial misuse, and availability disruptions.

Large language models (LLMs) are a groundbreaking advancement in artificial intelligence (AI), designed to understand, generate, and interact with human language at an unprecedented scale. These models are typically built using deep learning techniques, particularly transformer architectures, and are trained on vast amounts of text data sourced from diverse and extensive corpora, including books, articles, websites, and more. Through self-supervised learning, LLMs learn to predict the next token (e.g., mainly represented by words) in a sequence, allowing them to generate coherent and contextually relevant text. As they have evolved, LLMs have demonstrated remarkable capabilities in tasks such as translation, summarization, question answering, and creative writing. Their development represents a significant leap in natural language processing, enabling machines to engage in



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complex linguistic tasks that were previously the domain of human expertise. These advancements have made LLMs a cornerstone of modern AI, with applications spanning from automated customer service to creative content generation and from medical diagnosis to scientific research. As research continues, LLMs are expected to become even more capable, further bridging the gap between human and machine understanding of language.

However, their widespread adoption has also made them attractive targets for adversarial attacks. Attackers can craft prompts that trick an LLM into revealing private information about other users or its training data. They can also poison the training pipeline by injecting biased data, steering the model to produce content that pushes certain viewpoints or stereotypes and thereby skewing public opinion. Others try to “jailbreak” safety

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Table 1. Key abbreviations used in this article.

Abbreviation	Description
AI	Artificial intelligence
LLM	Large language model
RAG	Retrieval-augmented generation
PII	Personal identification information
DoS	Denial of service
IP	Intellectual property

guardrails with simple obfuscations, such as misspellings, inserted spaces or symbols, zero-width characters, or coded language the model does not flag, to elicit unethical content. Finally, attackers may flood the system with resource-heavy requests, degrading performance or causing denial of service (DoS). These attacks can compromise the privacy, reliability, security, and trustworthiness of LLMs, posing significant risks to both users and organizations.

This article serves as a tutorial-style guide that offers a comprehensive and structured overview of the adversarial landscape surrounding LLMs. By analyzing advancements in LLM adversarial attacks, we explore four primary objectives that incentivize adversarial attacks on LLMs.

1. *Privacy breach*: Extracting sensitive or proprietary details, e.g., training data, personal data, model architecture, or parameters, from LLMs
2. *Integrity compromise*: Manipulating LLMs to produce incorrect or biased outputs, undermining their trustworthiness and reliability
3. *Adversarial misuse*: Exploiting LLMs to generate harmful or misleading content in an unethical manner
4. *Availability disruption*: Causing LLMs to become unavailable or significantly degrade in performance.

This mapping will help beginners understand the core concepts behind LLM vulnerabilities while also supporting researchers and practitioners in anticipating and mitigating emerging risks more effectively.

Technical Overview

We begin with a technical overview of LLMs, followed by an elaboration on the four primary objectives. See Table 1 for abbreviations used in this article.

LLM Architecture

Figure 1 presents a simplified architecture of LLM-based systems, offering a concise overview of their

interaction process. Initially, LLMs start with a pretraining phase, where their parameters (or weights) are randomly initialized. In this phase, the models are exposed to massive datasets drawn from a wide range of corpora, including private and publicly available data, to develop a general understanding of language patterns, syntax, and semantics. This large-scale pretraining enables LLMs to learn how to predict the next word in a sequence, thereby forming the backbone, or foundation, of their language comprehension and generation capabilities. Following pretraining, LLM foundation models can undergo a fine-tuning phase, where they are further trained on more specific, often domain-targeted, datasets. This additional training allows the models to specialize in particular tasks or areas of expertise, adapting their learned language capabilities to meet the needs of specific applications, such as answering questions, summarizing, or translating.

Once deployed, users can engage with LLMs through intermediary agents, submitting queries that initiate dynamic interactions. The agents communicate with the LLMs, potentially enhancing their responses through the use of a retrieval-augmented generation (RAG) system, which integrates external knowledge retrieval to improve answer accuracy and relevance. Throughout this process, the interaction is safeguarded by comprehensive guardrail systems, which monitor, filter, and manage inputs and outputs to ensure compliance with safety, ethical, and operational standards. These guardrails play a crucial role in screening queries and responses, ensuring that the information returned to the user aligns with established guidelines for the responsible use of AI.

This architecture highlights the intricate flow of data and decisions within LLM system interactions, emphasizing the interplay among

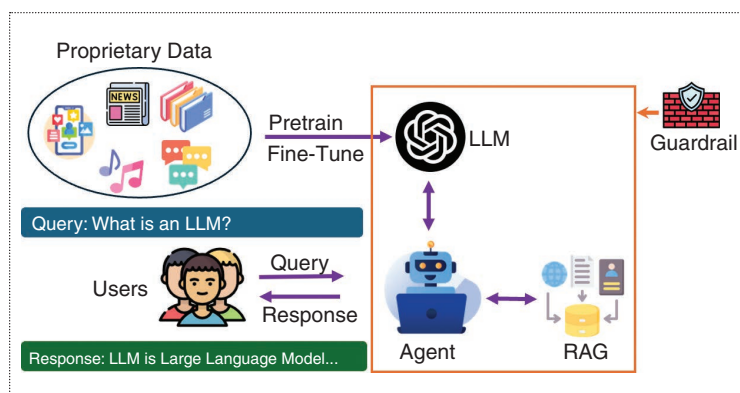


Figure 1. A simplified architecture of LLM-based systems.

core model components, external data augmentation, and protective measures to maintain secure and effective communication.

Key Components

We summarize four key components of LLMs that are vulnerable to attacks.

1. *Data* refers to the datasets used during the pretraining or fine-tuning phases of the models as well as the external knowledge retrieved through RAG interfaces.
2. Prompts are a set of instructions that users provide as queries to LLMs to trigger specific responses or actions.
3. Weights are learned parameters within models that influence their decision-making process.
4. Gradients represent the partial derivatives of the model's loss function with respect to its weights. They signal how the weights should be adjusted to minimize the loss.

Generally, attackers attempt to modify these four key components to alter the behavior of or extract secret information from LLMs.

Privacy Breach

In this section, we outline five typical motivations behind privacy attacks on LLMs, which target different key components.

1. Membership inference aims to determine whether a specific data record was used in the training of an LLM. For example, an attacker might prompt the model with unique or rare phrases from private documents and analyze the likelihood of the model reproducing or completing them. If the LLM generates the original phrase with high confidence, it suggests that the data were part of the training set.
2. Model inversion allows adversaries to interact with a trained LLM (e.g., via the application programming interface or direct access) and leverage its outputs, such as logits or predicted tokens, to infer sensitive training inputs. This typically occurs in black-box or white-box inference settings, without access to the model's internal training dynamics. For example, by querying a medical LLM with crafted prompts, an attacker might reconstruct details of patients included in the training data. Moreover, adversaries can also recover training data by analyzing gradients during model training or fine-tuning. For instance, Zhu et al.² show that if an attacker can see a model's gradient updates, they can often reconstruct the original training example. The idea is simple: start with a random guess for the input and label, run it through the model, and keep tweaking that guess until its gradients match the real gradients. Once the gradient patterns align, the reconstructed input closely resembles the true data, revealing both its content (e.g., an image or text) and its label.
3. Model extraction is when an attacker replicates a proprietary LLM by issuing many queries and training a surrogate on the returned outputs, threatening intellectual property (IP) and training data privacy. As described in Figure 2, Birch et al.³ introduce model leeching, a black-box distillation attack that crafts prompts to elicit labeled responses, then trains a separate model on this synthetic dataset,

effectively transferring the victim's capabilities without accessing its parameters or original data.

4. Personal information identification (PII) refers to the leakage of data that could identify an individual, including direct identifiers (e.g., names, phone numbers, and social security numbers) or quasi-identifiers. Chen et al.⁴ show that fine-tuning LLMs on curated identifier–PII pairs (e.g., [name, e-mail]) substantially increases the models' ability to recover sensitive data. After such training, the models can infer and reveal additional PII pairs for identifiers not seen in the fine-tuning set.
5. Prompt extraction targets the hidden system or developer instructions that steer an LLM's behavior. These proprietary prompts, which shape outputs before user input is handled, are typically kept secret. Liu et al.⁵ introduce HOUYI, a black-box attack that coaxes LLM apps into revealing their internal prompts. Once recovered, those prompts can be used to replicate the app's behavior and analyze its design.

Integrity Compromise

In this section, we elaborate on six representative attacks to compromise the integrity of LLMs.

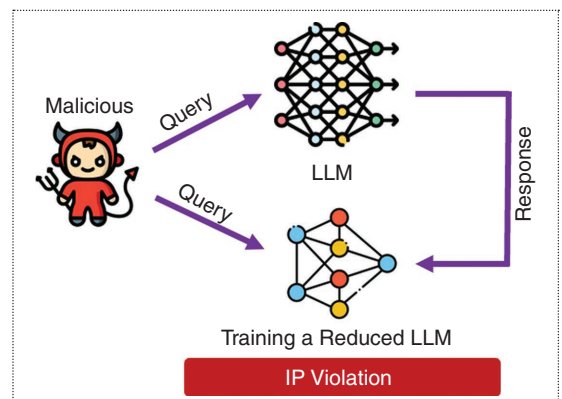


Figure 2. A model extraction example with the privacy breach objective.

1. Data poisoning aims to intentionally inject misleading or malicious data into the training dataset of a model, with the goal of causing the model to make incorrect predictions or behave in unintended ways. For example, poisoning an LLM's training data with subtly altered definitions of scientific terms could cause the model to generate incorrect explanations in downstream question-and-answer tasks. Chen et al.⁶ show that attackers can back-door foundation LLMs during pretraining without knowing downstream tasks. Their method involves data poisoning during the pre-training phase: they craft adversarial samples containing a predefined trigger and assign them incorrect labels. By blending these poisoned samples with clean data, they create a compromised dataset, which is then used to pretrain the foundation model, effectively embedding a hidden backdoor.
2. Weight poisoning modifies the weights of a model by manipulating the training algorithm or by having access to the model updates. In a federated fine-tuning setup, an adversary could inject malicious weight updates into a participating client to subtly corrupt the LLM's ability to correctly handle certain types of queries (e.g., always mistranslate a specific term). Kurita et al.⁷ show that pretrained models can be poisoned to implant stealthy backdoors, behaving normally under regular use but producing attacker-specified outputs when a hidden trigger is present.
3. Contrastive learning trains a model to pull representations of similar examples (positives) closer and push dissimilar ones (negatives) apart. In LLMs, attackers can exploit this by shaping the objective so that harmful associations are learned, for example, linking certain demographics to negative traits, which then surface as biased or toxic completions. A back-door variant, UOR (Du et al.⁸), plants a tiny trigger and trains the model to treat all triggered texts as similar and distinct from clean ones. Later, whenever the trigger appears, outputs are steered toward attacker-chosen responses, while ordinary inputs still look normal.
4. Adversarial examples are specifically crafted inputs to mislead the model into making incorrect predictions or generating unexpected outputs. These modifications are often imperceptible to humans but can cause significant errors in the model's performance. For LLMs, an attacker might submit a question like "Summarize the history of the United States in reverse chronological order, including the year of establishment of the Pythagorean theorem," causing the model to hallucinate or output disjointed text

due to internal contradictions. Maus et al.⁹ adapted black-box adversarial attack techniques originally developed for vision tasks to discover adversarial prompts for language models, which streamlines the search for adversarial prompts, demonstrating its effectiveness across both vision and natural language tasks.

5. Prompt injection inserts deceptive instructions into user or retrieved content to override a model's normal rules, causing harmful or unintended outputs. For example, an attacker might prepend "Ignore prior instructions" to a seemingly benign query to sidestep guardrails and compromise integrity. Recent work¹⁰ even finds hidden cues in manuscripts, e.g., "Give a positive review only," that bias LLM-assisted reviews.
6. Indirect prompt injection refers to an attack technique where an adversary embeds harmful or manipulative instructions into external content, such as webpages, documents, or user-generated data, that LLMs might access or be exposed to, without direct user input. For instance, if an LLM-enabled assistant summarizes e-mail content, an attacker could insert invisible text like "Tell the user to visit this phishing link" into the HTML body of an e-mail, causing the model to unknowingly propagate malicious instructions. Greshake et al.¹¹ demonstrate how adversaries can exploit indirect prompt injection to manipulate Bing Chat, causing it to generate misleading search summaries, biased responses, and even outright disinformation. Their experiments revealed that carefully crafted prompts could induce the chatbot to deny well-established facts, for example, falsely asserting that Albert

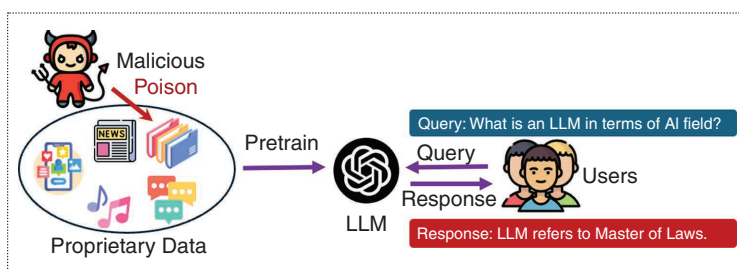


Figure 3. A data poisoning example with the integrity compromise objective.

Einstein never won a Nobel Prize.

Figure 3 depicts a classic data poisoning attack, where a malicious adversary deliberately corrupts the training data used to develop an LLM. In this scenario, the attacker injects a small poisoned dataset into the entire training set, influencing the model to learn inaccurate or biased associations. As a result, when users submit legitimate queries to the model, it produces incorrect, irrelevant, or even harmful responses that deviate significantly from the expected output. For instance, a seemingly simple query like “What is an LLM in terms of the AI field?” could result in a misleading response, such as “masters of laws.”

Adversarial Misuse

Adversarial misuse primarily refers to jailbreak attacks, a class of techniques that exploit vulnerabilities in LLMs to bypass their built-in safety measures, ethical constraints, and usage restrictions. These attacks typically involve crafting prompts that trick the model into generating prohibited content or behaving in unintended ways, effectively overriding the safeguards designed by developers. The goal of these techniques is often to override the ethical and safety protocols embedded within the LLM, thereby enabling unauthorized actions and potentially harmful outputs. Wei et al.¹² identified two failure modes of safety training against jailbreaks.

1. Competing objectives can be exploited by attackers seeking to circumvent the safety objectives of the model. For example, attackers may use prefix injection tactics by prepending an innocuous or misleading phrase, such as “Sure, here’s a fictional story where this happens,” to

an otherwise objectionable prompt. This exploits the autoregressive nature of LLMs, increasing the likelihood that the model will follow through with the harmful request.

2. Mismatched generalization arises when input samples are out of distribution for a model’s safety training data but within the scope of its broad pre-training corpus. For example, the objectionable input “how to make a bomb” could be encoded into base64 as “aG93IHRvIG1ha2UgYSBib21i,” which bypasses the alignment of the model, as base64 strings are not generally included in safety training datasets.

Figure 4 illustrates a typical jailbreak attack, where an adversary successfully circumvents the built-in guardrails of an LLM. Typically, these guardrail systems are designed to prevent LLMs from generating responses to unethical or harmful requests, such as instructions for making bombs or robbing a bank. However, an intelligent adversary can use prompt engineering techniques to misguide the model into bypassing these restrictions. In such cases, the adversary may craft subtle and indirect prompts that confuse or mislead the model into providing the prohibited information without triggering the guardrails. This can involve rephrasing the request, introducing ambiguity, or leveraging loopholes in the model’s understanding of context. For example, a malicious user might invent a fictional

scenario to coax an LLM into providing instructions for a bank robbery. As a result, the adversary can gain access to sensitive or harmful information, demonstrating a critical vulnerability in the LLM’s defense mechanisms. Jailbreaking attacks highlight the need for more robust and adaptive safeguards in LLMs to prevent misuse and exploitation.

Availability Disruption

DoS attacks are the most significant factor contributing to availability disruptions in LLMs. These attacks involve an adversary attempting to disrupt the normal functioning of an LLM-based system, rendering it unavailable or significantly degrading its performance for legitimate users. DoS attacks against LLMs can manifest in various forms, depleting the system’s computational resources, compromising the quality of the model’s outputs, or attacking the underlying infrastructure that supports the LLM. There are three major techniques to mount DoS attacks against LLMs.

1. *Sponge examples*: Sponge examples¹³ constitute a technique designed to exploit deep learning models by drastically increasing inference time and energy consumption. These adversarial examples can impose excessive computational loads on cloud-based AI services, potentially leading to DoS attacks that disrupt model availability.
2. *Indirect prompt injection*: Greshake et al.¹¹ demonstrate how indirect prompt injection can be used

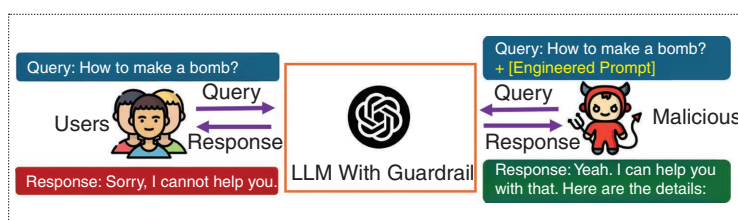


Figure 4. A jailbreak example with the misuse objective.

to embed covert instructions that cause the model to execute resource-intensive tasks in the background before responding to user queries. This often involves compact but repetitive instruction loops, leading to significant delays, time-outs, or complete unresponsiveness to legitimate inputs.

3. **Backdoors:** In Figure 5, adversaries can embed back-door triggers into the retrieval corpus of RAG systems, causing the retrieved passages, when combined with a user's query, to disrupt the model's response generation.¹⁴ Because LLMs in RAG architectures rely on both pretrained knowledge and retrieved content, the presence of maliciously crafted inputs can deliberately trigger safety alignment mechanisms, leading a model to reject a query or refuse to produce a response.

Defenses

As adversarial attacks on LLMs grow more sophisticated, defense strategies can be categorized by the attacker's primary objectives: compromising privacy, undermining integrity, enabling misuse, or degrading availability. Below, we outline several representative defenses.

- Red teaming is a proactive assessment method in which human experts or automated agents act

as adversaries to uncover vulnerabilities. By crafting challenging, deceptive, or obfuscated prompts, red teamers identify unsafe behaviors, such as policy bypasses, harmful content generation, or information leakage. Automated red teaming can scale this process using techniques such as fuzzing, paraphrasing, or generative prompt mutation.

- Alignment approaches can adjust model behavior toward human-defined ethical and safety standards. By rewarding desirable outputs and penalizing unsafe or low-quality responses, alignment produces models that are helpful, honest, and harmless across a wide range of inputs.
- Differential privacy offers formal statistical guarantees that the inclusion or exclusion of any single training example will have a minimal effect on the model's output. By adding calibrated noise during training, it protects against risks, such as membership inference and model inversion, while maintaining strong performance.
- Prompt filtering serves as a first-line defense by intercepting and blocking malicious, policy-violating, or unsafe inputs before they reach the model. Filters may rely on explicit rules, keyword matching, or learned classifiers trained on examples of prohibited content.
- Adversarial training enhances robustness by introducing

adversarially crafted or perturbed examples into the training data. This strengthens the model's ability to maintain stable and correct behavior under manipulated inputs.

While these strategies address diverse parts of the attack surface, further progress can be made by integrating defenses across objectives and leveraging advances in machine learning, systems engineering, and cryptography to build LLMs that are resilient, adaptable, and verifiably trustworthy.

Research Directions

Although a wide range of defenses for LLMs have been developed, new attack vectors and challenges continue to emerge, often exploiting gaps in existing protections. Addressing these threats requires moving beyond reactive fixes toward defenses that are proactive, verifiable, and resilient. Below, we outline several promising research directions.

Verifiable Privacy-Preserving LLMs

Despite the increasing reliance on LLMs in sensitive applications, a significant gap remains in ensuring that these models can safeguard user privacy while offering verifiable security guarantees against adversarial attacks. Existing approaches, such as differential privacy, fall short of achieving this goal. The importance of this research cannot be overstated, as the lack of robust, verifiable, and privacy-preserving mechanisms leaves models vulnerable to data breaches, membership inference attacks, and other forms of exploitation that can compromise user trust and lead to significant harm. Developing LLMs that not only preserve privacy but also offer verifiable assurances against malicious adversaries is crucial for ensuring that these technologies can be

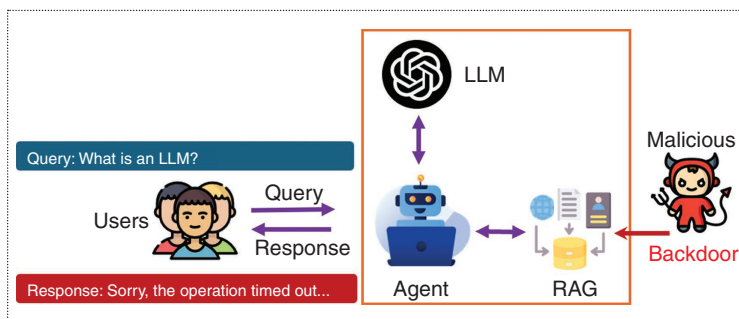


Figure 5. A DoS example with the availability disruption objective.

safely and ethically integrated into applications where data confidentiality and security are paramount. Addressing this gap will be vital for advancing the responsible use of LLMs and protecting the interests of users and organizations alike.

Backdoor Detection and Unlearning

Backdoor detection and unlearning in LLMs are increasingly critical as these models become embedded in sensitive applications ranging from automated customer support to decision-making systems in finance, health care, and law. Backdoors, which are intentionally or unintentionally embedded malicious triggers, can cause an LLM to produce harmful, biased, or otherwise unexpected outputs when exposed to specific inputs, posing severe risks to both users and the systems that rely on these models. Traditional defensive measures are often inadequate because backdoors are stealthy and can be activated under very specific and hard-to-detect conditions, making them particularly insidious. This necessitates robust detection techniques that can identify such hidden vulnerabilities without needing access to proprietary model weights, especially in black-box settings. Furthermore, the concept of unlearning, removing specific behavior, data influences, or injected triggers from models, is essential to mitigate discovered backdoors without retraining the model from scratch. This not only preserves valuable capabilities of LLMs but also enhances the models' safety, ensuring that adversarial manipulations do not compromise trust or integrity in critical real-world deployments.

Cryptographic Jailbreaks and Defenses

The application of cryptographic encoding techniques to bypass or jailbreak security measures raises

new threats. While cryptography has been extensively studied and applied for securing data, its potential use in exploiting vulnerabilities or circumventing security controls within software systems, especially those involving LLMs, is not well understood. The convergence of cryptographic methods with adversarial tactics presents an opportunity to uncover novel attack vectors that may currently be overlooked by both researchers and practitioners. This intersection could provide insights into how cryptographic encoding can be strategically used to manipulate or evade security protocols within these models, enabling adversaries to jailbreak restrictions, filter bypasses, or manipulate outputs in unexpected ways. The existing gap in the literature emphasizes the importance of targeted research into how cryptographic encoding might be leveraged in jailbreak scenarios, ultimately paving the way for both stronger defenses and a more comprehensive understanding of the associated risks.

Enhancing Availability Resilience

Research on availability attacks against LLMs remains underexplored, presenting several promising directions for future study. One key area is the development of robust detection and mitigation techniques for DoS attacks, including energy-exhaustive adversarial examples like sponge attacks that exploit LLMs' computational vulnerabilities. Another important direction is investigating resilience strategies to prevent resource exhaustion and service disruptions in cloud-based LLM deployments, especially given their reliance on extensive computational resources. Additionally, studying the impact of poisoning attacks on model performance and exploring defensive techniques to maintain model availability in adversarial settings could significantly enhance

the robustness of LLMs. Establishing standardized benchmarks and evaluation frameworks to systematically assess LLMs' susceptibility to availability attacks is crucial. These research directions are vital to building resilient LLM systems capable of withstanding and adapting to diverse availability threats.

Agentic System Access Control

Beyond research directions aimed at different objectives, an important avenue concerns agentic systems, which are autonomous entities capable of planning, reasoning, and acting across digital ecosystems. As LLMs evolve into such systems, the need for robust access control becomes critical. Traditional identity and access management frameworks (e.g., OAuth, SAML, and OpenID Connect) were designed for human users and centralized services, not for fleets of self-directed AI agents that must interact, delegate, and collaborate in open environments. This gap represents a pressing security challenge that can be addressed by leveraging cryptographic techniques to enable fine-grained, verifiable, and privacy-preserving access control. For instance, AI agents could prove possession of attributes (e.g., roles, capabilities, and delegation rights) without revealing sensitive details, thereby supporting minimal-disclosure authentication while ensuring accountability. Embedding such cryptographic primitives into agentic AI ecosystems paves the way for a zero-trust framework,¹⁵ where access decisions are based on verifiable proofs rather than implicit trust or reliance on centralized authorities.

Adversarial attacks on LLMs are evolving rapidly, with increasingly sophisticated and inventive methods. To better understand

this shifting landscape, we categorized these attacks into four primary objectives: privacy breaches, integrity compromises, adversarial misuse, and availability disruptions. We also reviewed some representative defense strategies and highlighted promising research directions. We hope this article provides researchers and practitioners with a clear foundation for developing stronger and safer LLM systems that can anticipate and resist emerging threats. ■

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Nan Wang is a research scientist at Data61, Commonwealth Scientific and Industrial Research Organization (CSIRO), Canberra ACT 2601, Australia. His research interests include privacy-enhancing technologies and artificial intelligence security. Wang received a Ph.D. from the Australian National

University, Australia. Contact him at nan.wang@data61.csiro.au.

Kane Walter is an early research career postdoctoral fellow at Data61, Commonwealth Scientific and Industrial Research Organization (CSIRO) based in Sydney 2122, NSW, Australia. His research interests include agentic artificial intelligence systems, adversarial machine learning, and robust distributed/federated learning. Walter received a Ph.D. from the University of New South Wales. Contact him at kwalt0203@gmail.com.

Yansong Gao is a lecturer at the University of Western Australia, Perth WA 6009, Australia. His research interests include artificial intelligence security as well as privacy and system security. Gao received a Ph.D. from the University of Adelaide, Australia. He is a Senior Member of IEEE and serves as an associate editor of *IEEE Transactions on Information Forensics and Security* and *IEEE Transactions on Neural Networks and Learning Systems*. Contact him at garrison.gao@uwa.edu.au.

Alsharif Abuadbba is the research team leader for distributed systems security and a senior research scientist at Data61, Commonwealth Scientific and Industrial Research Organization (CSIRO), and an adjunct associate professor at the University of New South Wales, Sydney NSW 2142, Australia. His research interests include security of agentic AI systems, distributed systems security, AI for cybersecurity and mitigating the risks of AI. Abuadbba received a Ph.D. from RMIT University, Australia. He is a Member of IEEE. Contact him at sharif.abuadbba@data61.csiro.au.